

Tarea 24

1.- Obtener la derivada de cada una de las siguientes funciones

a) $y = \sin 2x^2$

$y' = \cos 2x^2 (4x)$

Regla de la cadena

$y' = 4x \cos 2x^2$

b) $F(x) = \operatorname{tg} x^2$

$F'(x) = \sec^2 x^2 (2x)$

R.C.

$F'(x) = 2x \sec^2 x^2$

c) $g(x) = \sec \sqrt{x}$

$g'(x) = \sec \sqrt{x} \operatorname{tg} \sqrt{x} \left(\frac{1}{2\sqrt{x}} \right)$

R.C.

$g'(x) = \frac{1}{2\sqrt{x}} \sec \sqrt{x} \operatorname{tg} \sqrt{x}$

d) $y = \operatorname{ctg} \left(\frac{1}{x} \right)$

$y' = -\operatorname{csc}^2 \left(\frac{1}{x} \right) \left(\frac{-1}{x^2} \right)$

R.C.

$u(x) = \frac{1}{x} = x^{-1}$

$u'(x) = -x^{-2} = \frac{-1}{x^2}$

$y' = \frac{1}{x^2} \operatorname{csc}^2 \left(\frac{1}{x} \right)$

Nombre:

Día

Mes

Año

Folio

Tema:

$$e) F(x) = x^2 \sec x^2$$

$$f'(x) = x^2 \sec x^2 \overset{\text{P.C.}}{+} \sec x^2 (2x) + \sec x^2 (2x)$$

$$f'(x) = 2x x^2 \sec x^2 + \sec x^2 + 2x \sec x^2$$

$$f'(x) = 2x \sec x^2 (x^2 + 1) + \sec x^2$$

$$2.- \text{ Sea } f(x) = x \cos x$$

Obtener $f'(0)$

$$f'(x) = x \overset{\text{P.C.}}{[-\sin x (1)]} + \cos x (1)$$

$$f'(x) = -x \sin x + \cos x$$

$$f'(0) = -0 \overset{0}{\sin(0)} + \overset{1}{\cos(0)}$$

$$f'(0) = 1$$

3.- Derivar las siguientes Funciones

$$a) y = \arcsen x^2$$

$$y' = \frac{2x}{\sqrt{1-(x^2)^2}}$$

$$y' = \frac{2x}{\sqrt{1-x^4}}$$

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$$b) F(x) = \arctan \sqrt{x}$$

$$F'(x) = \frac{\frac{1}{2\sqrt{x}}}{1+(\sqrt{x})^2} = \frac{\frac{1}{2\sqrt{x}}}{1+x}$$

$$F'(x) = \frac{1}{2(1+x)(\sqrt{x})}$$

$$c) g(x) = \arctan x^3$$

$$g'(x) = \frac{3x^2}{1+(x^3)^2}$$

$$g'(x) = \frac{3x^2}{1+x^6}$$

$$d) y = \arccos \left(\frac{x}{3} \right)$$

$$y' = -\frac{\frac{1}{3}}{\sqrt{1-\left(\frac{x}{3}\right)^2}}$$

$$y' = \frac{\frac{1}{3}}{\sqrt{1-\frac{x^2}{9}}} = -\frac{\frac{1}{3}}{\sqrt{\frac{1}{1}-\frac{x^2}{9}}} = -\frac{\frac{1}{3}}{\sqrt{\frac{9-x^2}{9}}} = -\frac{\frac{1}{3}}{\frac{\sqrt{9-x^2}}{\sqrt{9}}}$$

$$= -\frac{\frac{1}{3}}{\frac{\sqrt{9-x^2}}{3}} = -\frac{1}{\sqrt{9-x^2}}$$

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Tema:

e) $F(x) = \arcsin 2x$

$$F'(x) = \frac{2}{2x\sqrt{2x^2-1}}$$

$$F'(x) = \frac{1}{x\sqrt{4x^2-1}}$$

4.- Sea $y = \arctg \sqrt{x}$

Obtener $F'(1)$

$$y' = \frac{\frac{1}{2\sqrt{x}}}{1+(\sqrt{x})^2} = \frac{\frac{1}{2\sqrt{x}}}{1+x} = \frac{1}{2(1+x)\sqrt{x}}$$

$$F'(1) = \frac{1}{2(1+1)\sqrt{1}} = \frac{1}{4}$$